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ORIGINAL ARTICLE

Experimental analysis of tribological performance of palm oil blended with hexagonal boron nitride nanoparticles as an environment-friendly lubricant

Mohd Fadzli Bin Abdollah[1,2] · Hilmi Amiruddin[1,2] · Arina Dayana Jamallulil[1]

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Abstract
Expanded worries about the ecological harm brought about by mineral oil-based lubricants have made a developing worldwide trend of advancing vegetable oil as base oil. Along these lines, this paper presents the tribological performance of palm oil blended with hexagonal boron nitride (hBN) nanoparticles as an environment-friendly lubricant. The hBN nanoparticles were 70 nm and were blended with the palm oil at varying concentrations that ranged between 0 and 0.5 vol.%. The samples were blended by using the sonication technique, with the help of an ultrasonic homogenizer. Thereafter, the tribological test was conducted using a four-ball tribometer as indicated by engineering standard testing method (ASTM D4172-94). The surface morphology and chemical composition of the worn surfaces were observed using the scanning electron microscopy (SEM) installed with energy-dispersive X-beam spectroscopy (EDX). Results showed that when hBN nanoparticles were blended with palm oil, the unsteady-state coefficient of friction (*COF*) was promptly changed to a steady-state condition during the running-in period. However, the higher concentration of hBN nanoparticles causes higher *COF*. Furthermore, 0.1 vol.% of the hBN nanoparticles in palm oil exhibited lower *COF* and wear rate compared with the palm oil alone and commercially available engine oil SAE 15W40.

Keywords Tribology · Palm oil · Bio-lubricant · Hexagonal boron nitride

1 Introduction

A majority of the lubricants that are currently available in the market include petroleum-based mineral oils. These petroleum-based oils showed an adverse environmental effect as they are non-biodegradable and toxic, thereby requiring a higher waste management budget. The bio-based oils are derived from the vegetable oils and have been considered to be a better substitute for the petroleum-based oils, as they showed many environmental benefits with regard to their renewability, biodegradability, toxicity, and excellent performance in various applications [11].

Though the bio-based oils show many environmental advantages, some of the vegetable oils like palm oils display an imperfect oxidation and thermal stability, get solidified at lower temperatures, show poor fluid flow behaviour, and consist of double bonds in their fatty acid carbon backbone, thereby leading to a higher wear rate [12]. These limitations displayed by the vegetable oil could be overcome by using additive mixtures and also by genetically and chemically modifying the base oil [14].

Other than the technical issue, palm oil has related to the pulverization of the world's valuable rainforests according to WWF report [1]. The advancement of new

✉ Mohd Fadzli Bin Abdollah
mohdfadzli@utem.edu.my

[1] Fakulti Kejuruteraan Mekanikal, Universiti Teknikal Malaysia Melaka, Hang Tuah Jaya, 76100 Durian Tunggal, Melaka, Malaysia

[2] Centre for Advanced Research on Energy, Universiti Teknikal Malaysia Melaka, Hang Tuah Jaya, 76100 Durian Tunggal, Melaka, Malaysia

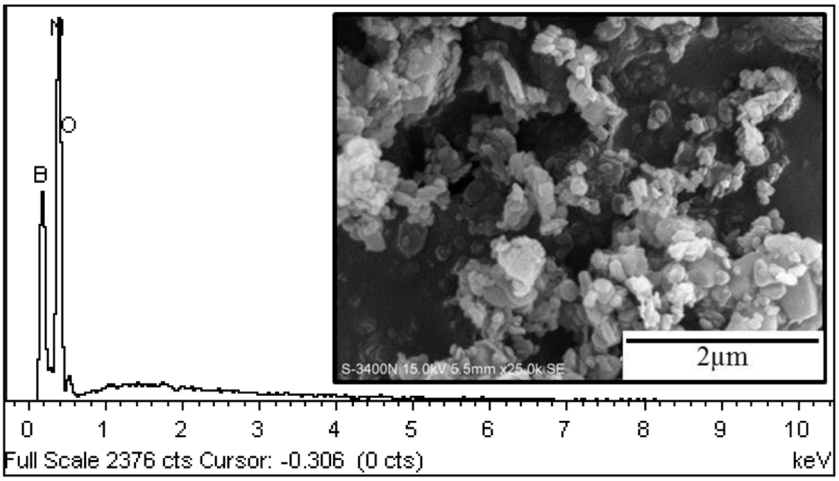


Fig. 1 EDX spectrum and SEM micrograph for the hBN nanoparticle

oil palm plantations, combined with smallholders extending their homesteads to satisfy the rising need for palm oil, has brought about huge deforestation. Nonetheless, palm oil is an unfathomably proficient yield, creating more oil per land zone than some other equal vegetable oil crop. Globally, palm oil supplies 35% of the world's vegetable oil demand on just 10% of the land. Internationally, palm oil supplies 35% of the world's vegetable oil request on only 10% of the land. To get a similar measure of elective oils like soybean or coconut oil, you would require anything somewhere in the range of 4 and multiple times more land, which would move the issue to different pieces of the world and compromise various natural surroundings and species. Moreover, palm oil is an essential crop for the gross domestic product of rising economies and there are many smallholder ranchers who rely upon delivering palm oil for their occupation. Boycotting palm oil is not generally the appropriate response yet requesting more activity to handle the issues and go further and quicker is.

Table 1 Physical and chemical properties of the hBN nanoparticles

[a]Properties	hBN nanoparticles
Appearance	White powder
Chemical formula	BN
Crystal structure	Hexagonal
Melting point, °C	3000 dissociates
Average diameter particle size, nm	70
Density, kg m^{-3}	2.3
Hardness, HRC	40
Maximum use temperature in air, °C	1000
Thermal conductivity, Wm^{-1} K^{-1}	27
Thermal expansion coefficient @25–1000 °C	1×10^{-6}/°C (parallel to press dir.)

[a] From manufacturer

Palm oil can be produced more sustainably, and things can change. The Roundtable of Sustainable Palm Oil (RSPO) was developed in 2004 because of expanding worries about the effects palm oil was having on the earth and society. The RSPO has a creation standard that sets best works on delivering and sourcing palm oil, and it has the upfront investment of the greater part of the worldwide business. Palm oil is Nature's Gift to Malaysia and Malaysia's to the World. As expressed by the Malaysian Palm Oil Council, Malaysia right now represents 28% of world palm oil creation and 33% of world fares. Whenever assessed different oils and fats delivered in the nation, Malaysia represents 9.5% and 19.7% of the world's entire generation and fares of oils and fats, respectively. Being probably the most significant maker and exporters of palm oil and palm oil items, Malaysia has a considerable task to carry out in satisfying the developing worldwide requirement for oils and fats sustainably. Malaysia currently delivers 45.3% of all the RSPO-certified palm oil in the world market. Representing the Malaysian palm oil industry's commitment towards biodiversity is the Malaysian Palm Oil Wildlife Conservation Fund, which was propelled in 2006 with a RM10mil award from the Federal Government and another RM10mil from the industry, to help condition and biodiversity protection endeavors [8].

The green solid lubricants like the hexagonal boron nitride (hBN) could form very stable compounds which

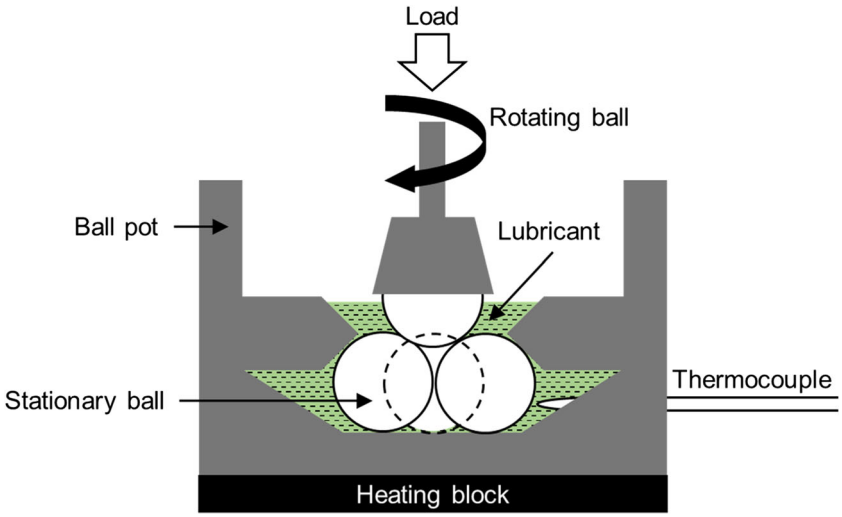


Fig. 2 Schematic diagram of the four-ball tribometer

were non-toxic, safe to handle, showed excellent thermal stability, and had no limitation with regard to their operational usage. The hBN nanoparticles were available as a lamellar powder that could form a layered crystalline structure, with weaker Van der Walls interactions occurring between every single layer, and strong covalent bonds occurring within the layer, thereby showing lower friction and decreasing the wear and contact surface [9]. Due to their unique characteristics, the hBN particles were used as an attractive performance-improving alternative to the inorganic solid lubricating molecules. Several researchers have investigated the effect of the hBN nanoparticle additives on their dispersion stability [5, 10]. They found that the stability of the hBN nanoparticles in SAE 15W-40 diesel engine oil increases in the alkaline region or using sodium benzene dodecyl sulfonate surfactant. The suspension was stable for 60 days, as compared with the suspension in the acidic region or sodium cholate surfactant. Meanwhile, other researchers found that the presence of hBN nanoparticles in conventional engine oil reduced the friction and wear performance of the sliding components [2, 4, 7, 15].

Table 2 Mechanical properties of the ball material

[a]Properties	Ball bearing (carbon chromium steel)
Hardness (H), HRC	61
Density (ρ), g/cm^3	7.79
Surface roughness (R_a), μm	0.022

[a] From laboratory measurements

In order to bridge the gap, this study investigates the tribological performance of the palm oil blended with hBN nanoparticles as an environment-friendly lubricant.

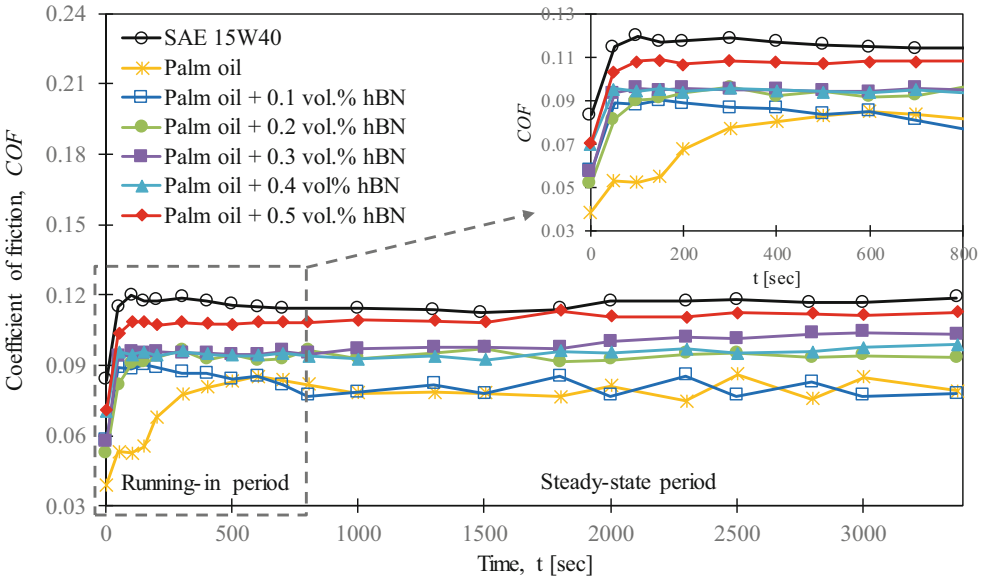
2 Methodology

2.1 Preparation of nano-based bio-lubricant

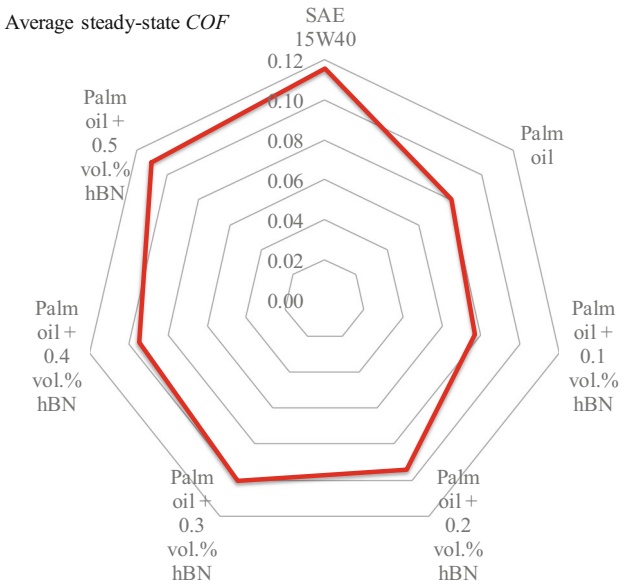
In this study, the nano-based bio-lubricant samples were prepared by dispersing 70 nm hBN nanoparticles in palm oil. The EDX spectrum and SEM micrograph of hBN nanoparticles is shown in Fig. 1. The image showed that the hBN nanoparticles had a spherical shape. These particles were dispersed in the oil by sonication using the ultrasonic homogenizer (50% amplitude at an interval of 0.5, 30 mins). hBN nanoparticle concentration that was dispersed in the palm oil varied between 0 and 0.5 vol.%. Table 1 presents the physical and chemical properties of hBN nanoparticles used in the study.

2.2 Tribological test

Tribological testing was conducted for determining the friction coefficient and wear rate for the contact surfaces, with the help of a four-ball tribometer, as shown in Fig. 2. This test procedure was based on the ASTM D4172-94 (2016) standard. For this test, the speed, load, time, and temperature values were set at 1200 rpm, 392 N, 3600 s, and 75 °C, respectively. This test was further repeated thrice for ensuring the reliability and accuracy of the



(a)



(b)

Fig. 3 Variations in the *COF* of the palm oil blended with the hBN nanoparticles: **a** dynamic *COF* and **b** average steady-state *COF*

results. The four-ball tribometer included three carbon chromium steel balls (12.7-mm diameter) that were covered by the lubricant for analysis. A 4th steel ball, with the same diameter (called the top ball), was held in the special collet within the spindle and was rotated using an AC motor. This top ball was rotated and was in constant contact with the three fixed steel balls which were immersed with the sample lubricant. The instrument recorded the coefficient of friction (*COF*) with the help of a data terminal processing system. Furthermore, the volume of the ball's worn material was estimated geometrically from the wear scar diameter that was measured using a digital microscope. Table 2 presents the mechanical properties of all the balls used in the test.

2.3 Surface morphology observation

The surface morphology of the worn surfaces was observed using the scanning electron microscopy (SEM), whereas their chemical composition was studied using the energy-dispersive X-ray (EDX) spectroscopy.

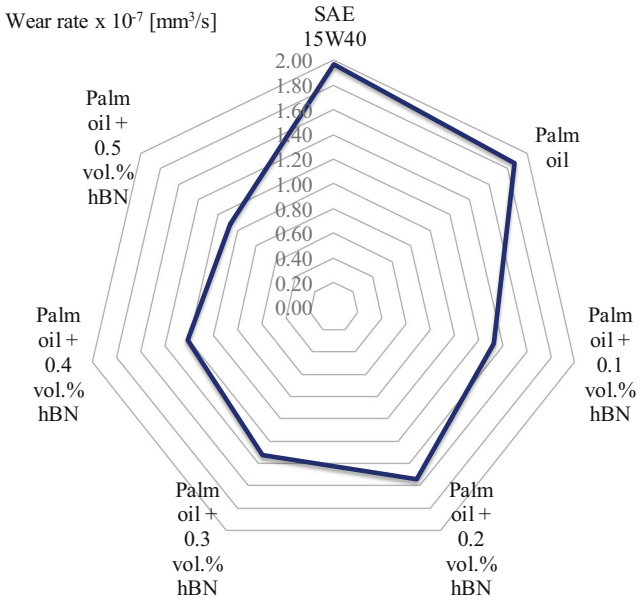


Fig. 4 Variation in the wear rates of the palm oil blended with hBN nanoparticles

3 Results and discussion

Figure 3 shows the variation of *COF* for the nano-based bio-lubricant samples. The data of commercially available engine oil SAE 15W40 is deployed for comparison purposes. All *COF* values for the nano-based bio-lubricant samples ranged between 0.08 and 0.10. The results showed that 0.1 vol.% concentration of the hBN nanoparticles that were dispersed in palm oil showed the lowest *COF* value compared with other samples and showed a 33% decrease in comparison with the conventional SAE petroleum-based oil. However, the *COF* value was similar to the control samples, i.e., sample without any hBN nanoparticles. It was also seen that hBN nanoparticles helped in developing a thin lubricating film, which altered the type of friction from the sliding type to rolling. Similar results were reported by Abdullah et al. [3], who noted that the inclusion of hBN nanoparticles in engine oil reduced the contact area. Reeves et al. [12] showed that dispersion of the

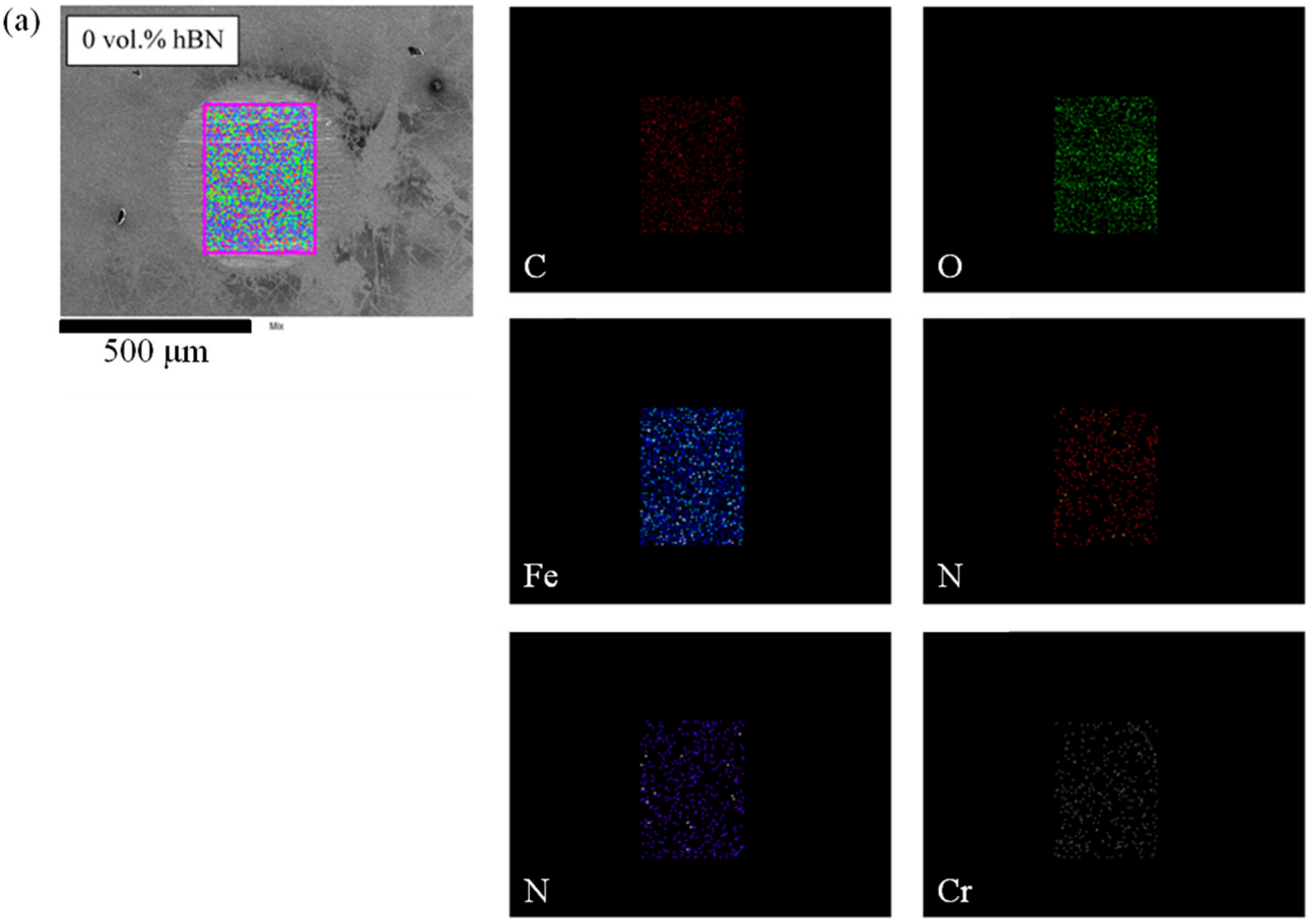


Fig. 5 The EDX elemental mapping of the worn steel surfaces that were lubricated with **a** palm oil (0 vol.% hBN), **b** nano-based bio-lubricant (0.3 vol.% hBN), and **c** nano-based bio-lubricant (0.5 vol.% of hBN)

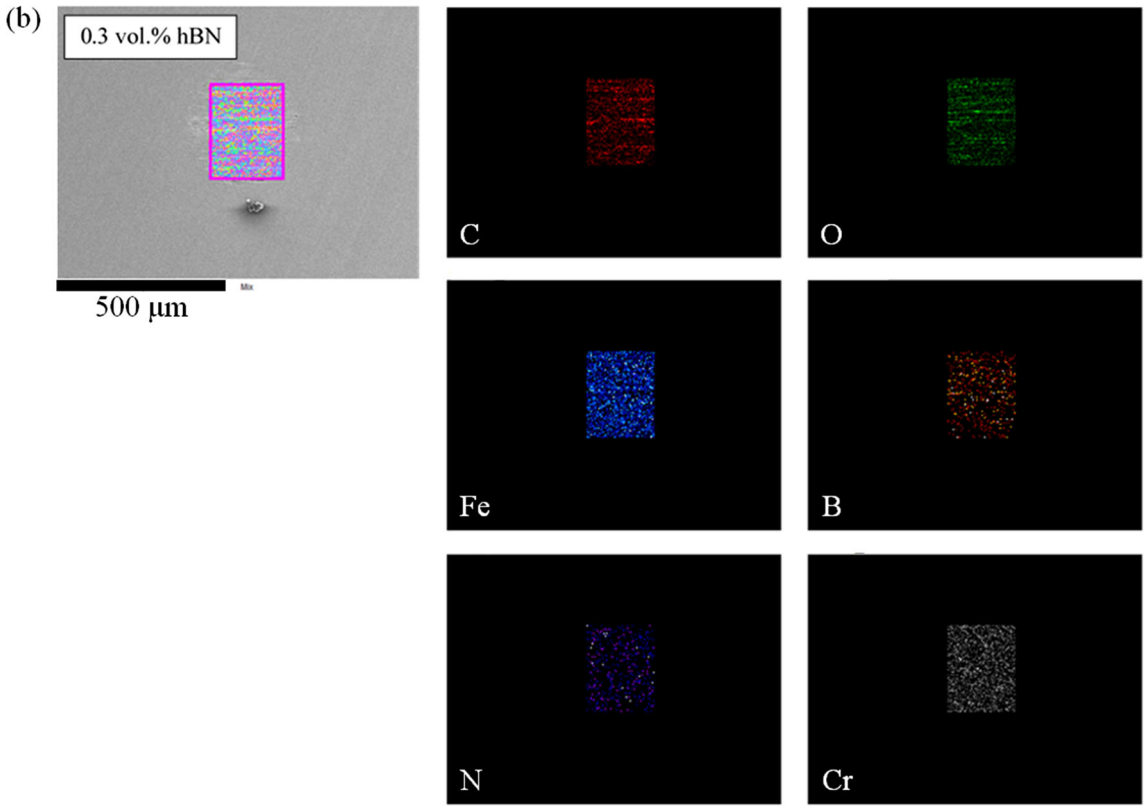


Fig. 5 (continued)

hBN nanoparticles in oil improved the lubrication, as they aligned themselves in a similar direction as the relative motion of the solid bodies, allowing them to slide smoothly over each other. On the other hand, the high hBN nanoparticle concentration in palm oil could increase the *COF* value, significantly. This was attributed to the high concentration of the hBN nanoparticles in palm oil, which caused agglomeration that further increased the shear strength and led to a high *COF*.

The small insert that is shown in Fig. 3a showed the running-in period, wherein the *COF* varied between its initial value and relatively stable value. The results showed that the running-in period for the commercially available engine oil SAE 15W40 and the palm oil blended with hBN nanoparticles, i.e., 100 s, was lesser than the palm oil sample without the hBN nanoparticles, i.e., 400 s, and remained stable throughout the friction test. Thus, it was concluded that the hBN nanoparticles were an effective additive to oil, which helped in stabilizing the *COF* during the initial operation. Many researchers in the past 100 years showed that an appropriate running in could significantly increase the lifespan of the engineering parts, though the phenomenon was not adequately understood earlier. Running in is described as a rapid process, wherein the tips of the asperities were plastically deformed in the initial few loading cycles. The valley features did not undergo any change, but the asperity tips flattened, thereby negatively skewing the distribution of the asperity height. The earlier experiments that investigated the running in of the lubricated surfaces referred to the gradual decrease in the roughness as the wear process and included only the plastic deformation of the asperity contacts [6]. This indicated that the thickness of the lubricant or the viscosity of the palm oil blended with hBN nanoparticles tends to increase, while pressure spikes or a lesser severity were observed at every "micro-contact" because of increasing conformity when the asperities became more rounded.

Figure 4 depicts the fact that the dispersion of the hBN nanoparticles in the palm oil could decrease the wear rate; however, the *COF* was increased with an increasing nanoparticle concentration. A tribofilm was formed because of the reaction between the fatty acids and the metal oxide

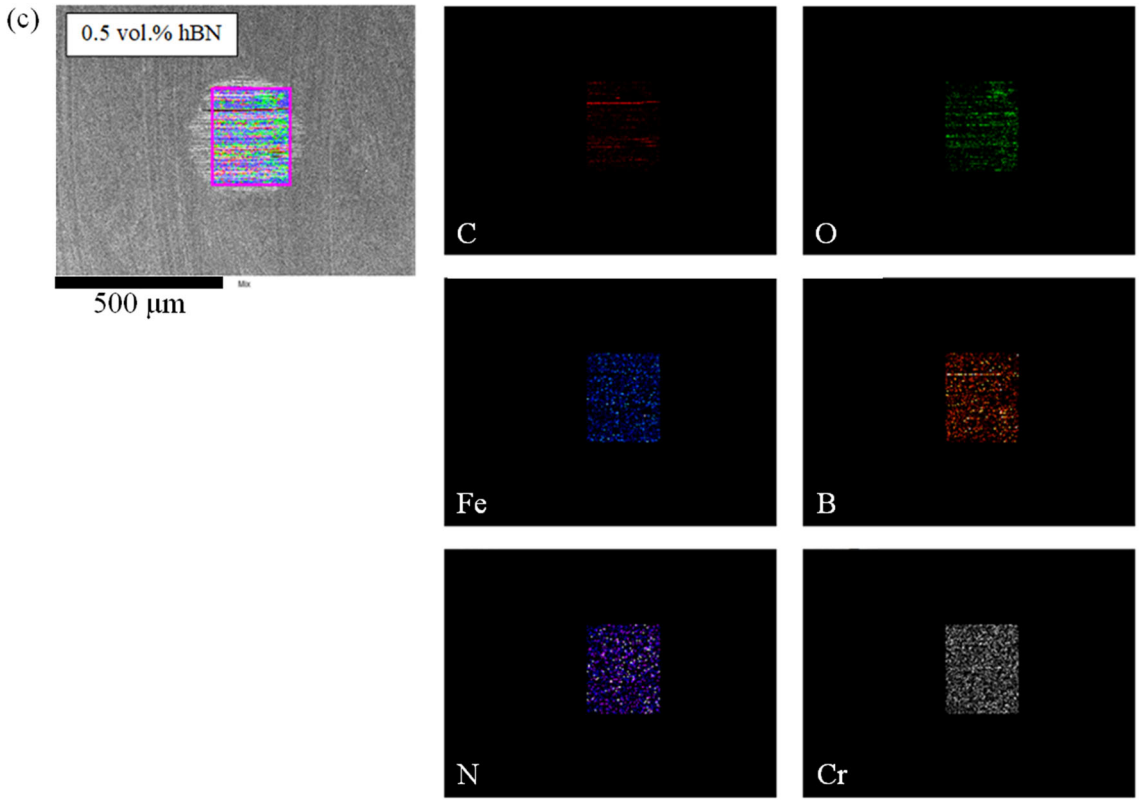


Fig. 5 (continued)

layer. This was attributed to the presence of the oxygen (O) and boron (B) molecules (Fig. 5) present on the worn surface of the steel balls that were lubricated with the nano-based bio-lubricant. This potentially reduced the shear forces acting on the steel ball interface, thereby decreasing the wear rate.

Figure 6 shows the formation of parallel shallow and deep grooves on the worn surface of the steel balls that were lubricated only with the palm oil (without hBN nanoparticles). The micrographs showed that this palm oil film without the hBN nanoparticles was entirely destroyed, which led to severe and abrasive wear because of the polished wear mechanisms. Furthermore, smooth worn surfaces having fewer grooves were seen on the steel ball surfaces that were lubricated with the nano-based bio-lubricant. These grooves on the smooth worn surfaces were attributed to the debris that was formed due to the bearing surface of the detached steel balls [13].

For better understanding, Fig. 7 presents the frictional wear mechanisms involved in the contact surfaces that were lubricated using the nano-based bio-lubricant. The low volume concentration of hBN nanoparticles in palm oil induces a rolling effect, which would reduce the shear strength and significantly decrease the *COF* and wear rate. Further addition of the nanoparticles increases the formations of agglomeration and which prevents the nanoparticles move freely because of the high shear strength. This mechanism causes an increase in *COF*, meanwhile maintaining or lower the wear rate due to the non-direct contact phenomena between surfaces.

4 Conclusions

Based on the abovementioned experimental results, it could be concluded that the dispersion of the hBN nanoparticles in the palm oil promptly changed the unsteady-state *COF* to a steady-state condition during the running-in period. Furthermore, the hBN nanoparticle additives concentration also affected the palm oil's tribological

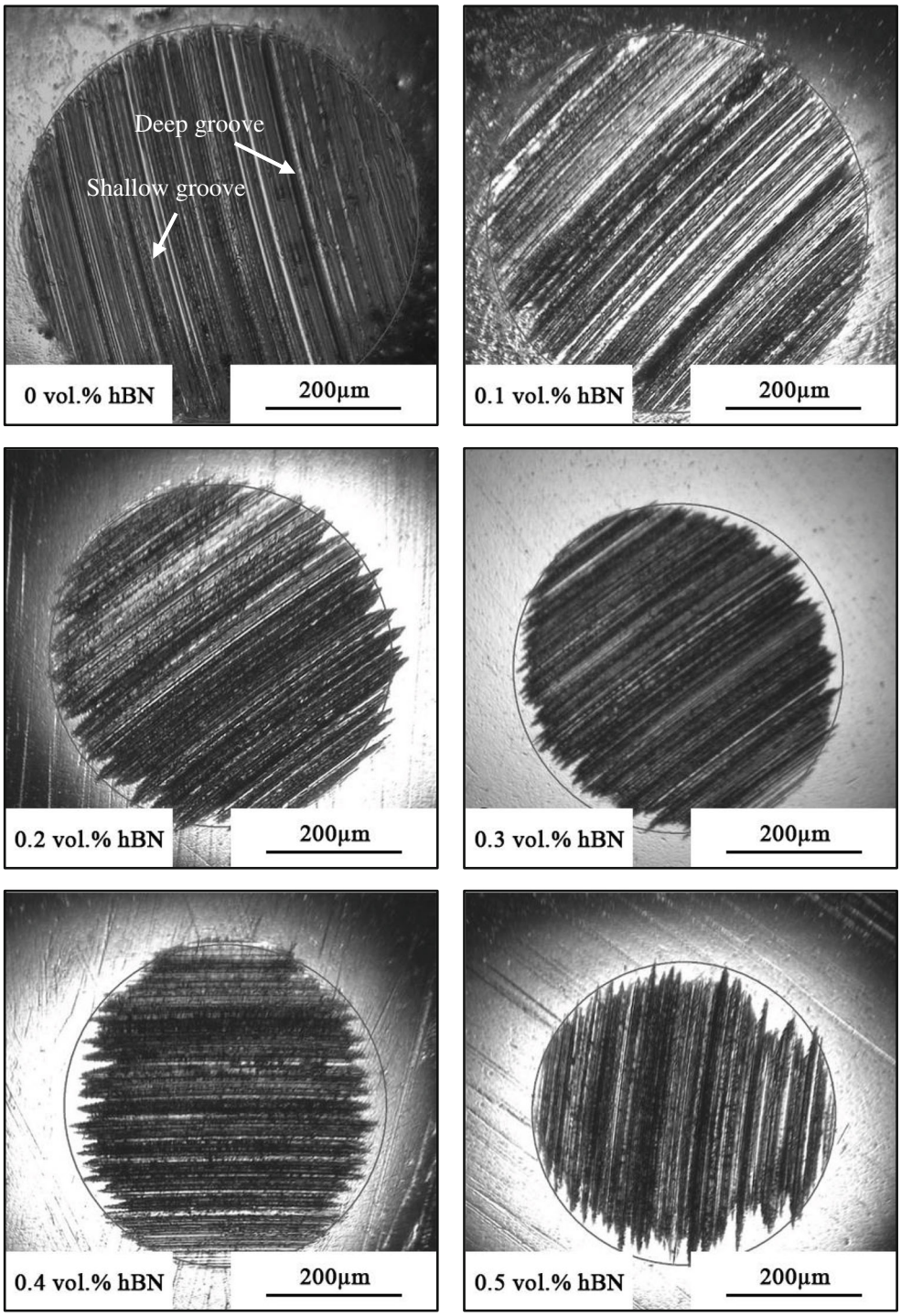


Fig. 6 Micrographs depicting the worn steel surfaces of the balls that were lubricated with palm oil (0 vol.% hBN) and nano-based bio-lubricant (0.1 to 0.5 vol.% hBN)

performance. High hBN nanoparticle concentration increased the *COF* and decreased the wear rate due to the agglomeration and tribofilm formation that took place. However, the *COF* and wear rate values of 0.1 vol.% of the hBN nanoparticles were still lower than the values displayed by the palm oil alone and commercially available engine oil SAE 15W40, as the nanoparticles exhibited a rolling effect.

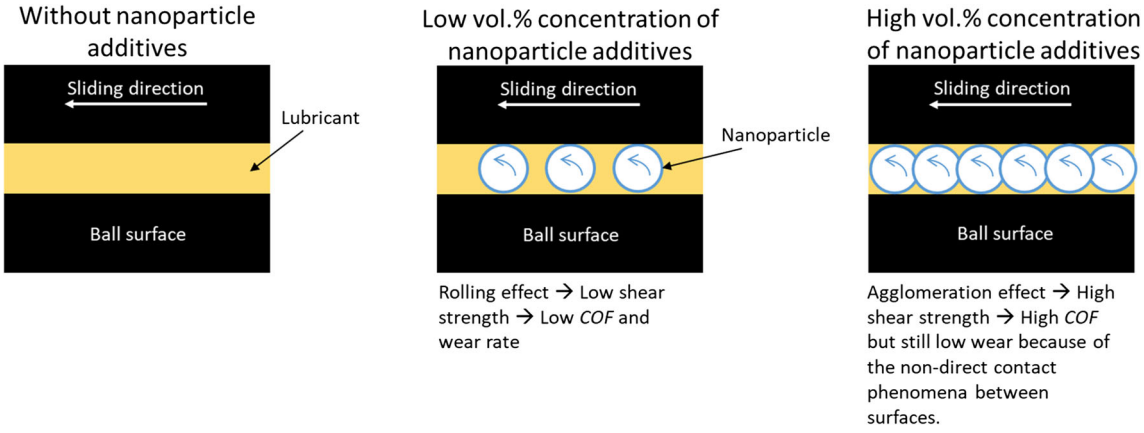


Fig. 7 Illustration of the frictional wear mechanisms of the mating surfaces lubricated with nano-based bio-lubricant

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